Cyber Attacks Explained
Device Evasions

This is the last installment of the year-long series on cyber attacks. As we know, network infrastructure is protected by security devices such as routers and firewalls. So far, it was believed that such devices were enough to secure the perimeter. This belief has been proved wrong by a technique called device evasion, which is a new trend in attacks in the cyber security world. This article explores this technique in detail, and also covers methods to prevent IT infrastructure from such attacks.

By definition, evasion is the process of avoiding or bypassing an object or a situation. In technical terms, evasion is a technique by which an attacker bypasses a security system in the cyber security space. The system may typically consist of routers, firewalls, network switches and intrusion detection devices. As we all know, routers segregate networks; firewalls block unwanted IP address and TCP port communications, whereas intrusion detection devices add a layer of intelligence based on anomaly detection techniques. A few years back, when these techniques were implemented network administrators could safely say that their infrastructure was secure. However, the laws of human nature tell us that a thief is always a step ahead of the cops! While these devices seem to thwart the attacks for some time, they have made cyber criminals more aggressive and have prompted them to come up with ways to penetrate and break down the security perimeter. Attackers use evasion methods in order to steal data, disrupt IT networks or plant software exploits.

As shown in Figure 1, a typical secure infrastructure contains at least a router, with a switch and a firewall also incorporated. To learn more about device evasions, let's concentrate on these three devices, and cover a bit about IDS systems too. Let's also learn how each of these devices can be broken into, and finally, how to secure them to protect